

The Global Thermostat: Where Should We Set It?

By Brendan Clark

Climate intervention techniques to counteract warming from anthropogenic climate change, such as the use of aerosols to reflect sunlight, could be a potential source of global conflict, reports a study published in *Nature Food*. The findings show that different nations would prefer different global temperatures to produce the most calories from crops, potentially leading to disagreement and conflict.

Brendan Clark and colleagues analyzed global climate and crop model data that simulated various climate change and climate intervention scenarios. The research shows that higher latitude nations tend to produce the most calories from crops under unmitigated climate change, while midlatitude nations produce the most calories under moderate climate intervention, and lower latitude nations produce the most calories from crops under large amounts of stratospheric aerosol climate intervention.

Even though, on a global scale, climate intervention tends to increase crop production, different nations would prefer different levels of intervention and different temperatures, with some preferring no intervention at all. This makes choosing one temperature for everyone difficult.

“Not one of the 11 climate change or climate intervention scenarios we analyzed benefits everyone,” said Brendan Clark, a doctoral student at Rutgers University. “Nations may have different ideas of what constitutes an optimal global temperature, which could lead to conflicts. It would be like people fighting over the thermostat in a house, but on a global scale.”

The study focuses on impacts just to national crop production, but the authors highlight that including other impacts, such as to ecology and other natural ecosystems, would only complicate the issue further.

“Are we willing to live with all these potential impacts to have less global warming? That’s the question we’re trying to ask here,” said Alan Robock, a Distinguished Professor of Climate Science in the Department of Environmental Sciences at SEBS, and a co-author of the study. “We’re trying to quantify each of the potential risks and benefits so we can make informed decisions in the future.”

The authors state that more research is needed to update current models as well as using other climate and crop models to help reduce uncertainties.

Other scientists on the study included Sam Rabin, Simone Tilmes and Jadwiga Richter of the National Center for Atmospheric Research; and Daniele Visoni of Cornell University.